SOFTWARE ENGINEERING BCSE301P

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1. Project Title

CROWD- SOURCE CIVIC ISSUE REPORTER AND RESOLUTION SYSTEM

2. Problem Statement

In urban areas, citizens frequently encounter civic issues such as potholes, garbage accumulation, sewage leaks, and broken infrastructure. However, reporting these issues is often inefficient due to:

- Lack of awareness about the correct department
- Time-consuming manual complaint processes
- Poor tracking and delayed response

As a result, many issues remain unresolved, affecting public safety, hygiene, and quality of life.

3. Objectives

The main objectives of this project are:

- To develop an AI-powered system that automatically detects civic issues from images
- To generate structured complaints including title, category, department, and description
- To provide a user-friendly platform for citizens to report issues easily
- To implement secure authentication for users and admins
- To store and manage complaints using a database
- To create an admin dashboard for efficient complaint management

4. Methodology / Approach

The system follows a full-stack AI-based architecture integrating frontend, backend, database, and AI services.

Step 1: User Interaction

- Users register and log in (Citizen/Admin roles)
- Citizens upload an image of a civic issue

Step 2: Image Processing

- The uploaded image is resized and converted into Base64 format using the image processor

Step 3: AI Analysis

- The image and user input are sent to a large language model via the Groq API
- A structured system prompt ensures consistent output:
    - Title
    - Category
    - Department
    - Description

Step 4: Data Storage

- Complaint details are stored in MongoDB
- Passwords are securely hashed using bcrypt

Step 5: Output Display

- AI-generated complaint is displayed to the user
- Complaint is saved for admin access

Step 6: Admin Dashboard

- Admin can view complaints categorized by department
- Complaints are displayed with metadata such as status, timestamp, and images

5. Work Completed So Far (50% Implementation)

The following components have been successfully implemented:

1. Authentication System

- User registration and login functionality implemented
- Role-based access (Citizen and Admin)
- Password hashing using bcrypt for security

2. AI Integration

- Integrated Groq API with LLaMA model
- Implemented structured prompt-based AI response generation
- AI can:
    - Analyze uploaded images

    - Detect issue type
    - Assign department
    - Generate complaint description

3. Image Processing Module

- Image resizing and compression implemented
- Conversion to Base64 format for API transmission

4. Database Integration (MongoDB)

- User data storage
- Complaint data storage
- Department-based retrieval of complaints

5. Citizen Dashboard

- Upload image functionality
- Optional text input for additional details
- AI-generated complaint preview
- Clean UI using Gradio with custom CSS

6. Admin Dashboard

- Department-wise complaint viewing
- Complaint cards with:
    - Title
    - Description
    - Status
    - Timestamp
- Image gallery for visual inspection

7. Complaint Processing Pipeline

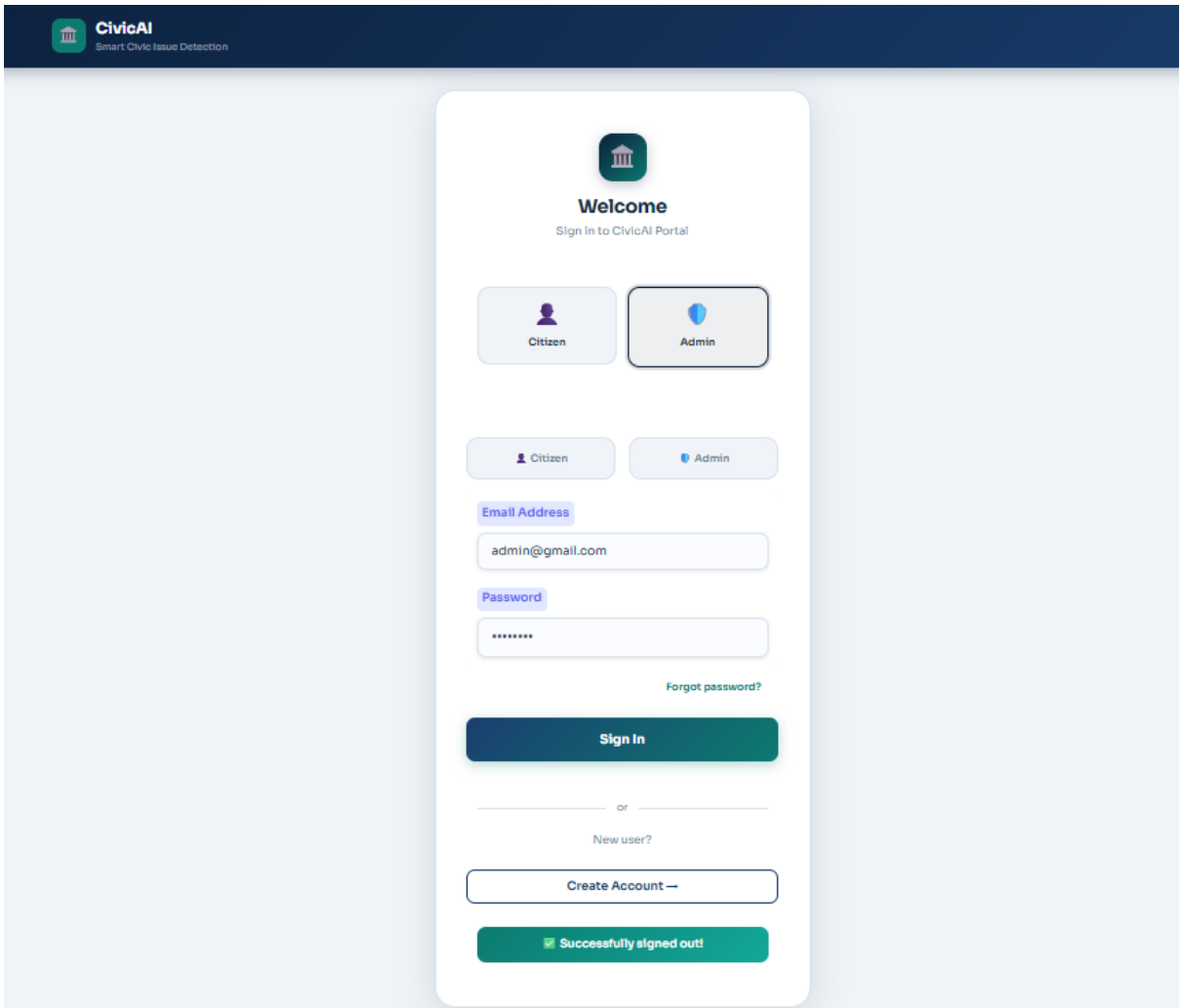
- End-to-end workflow:
    - Upload → AI Analysis → Parsing → Storage → Display

8. UI/UX Design

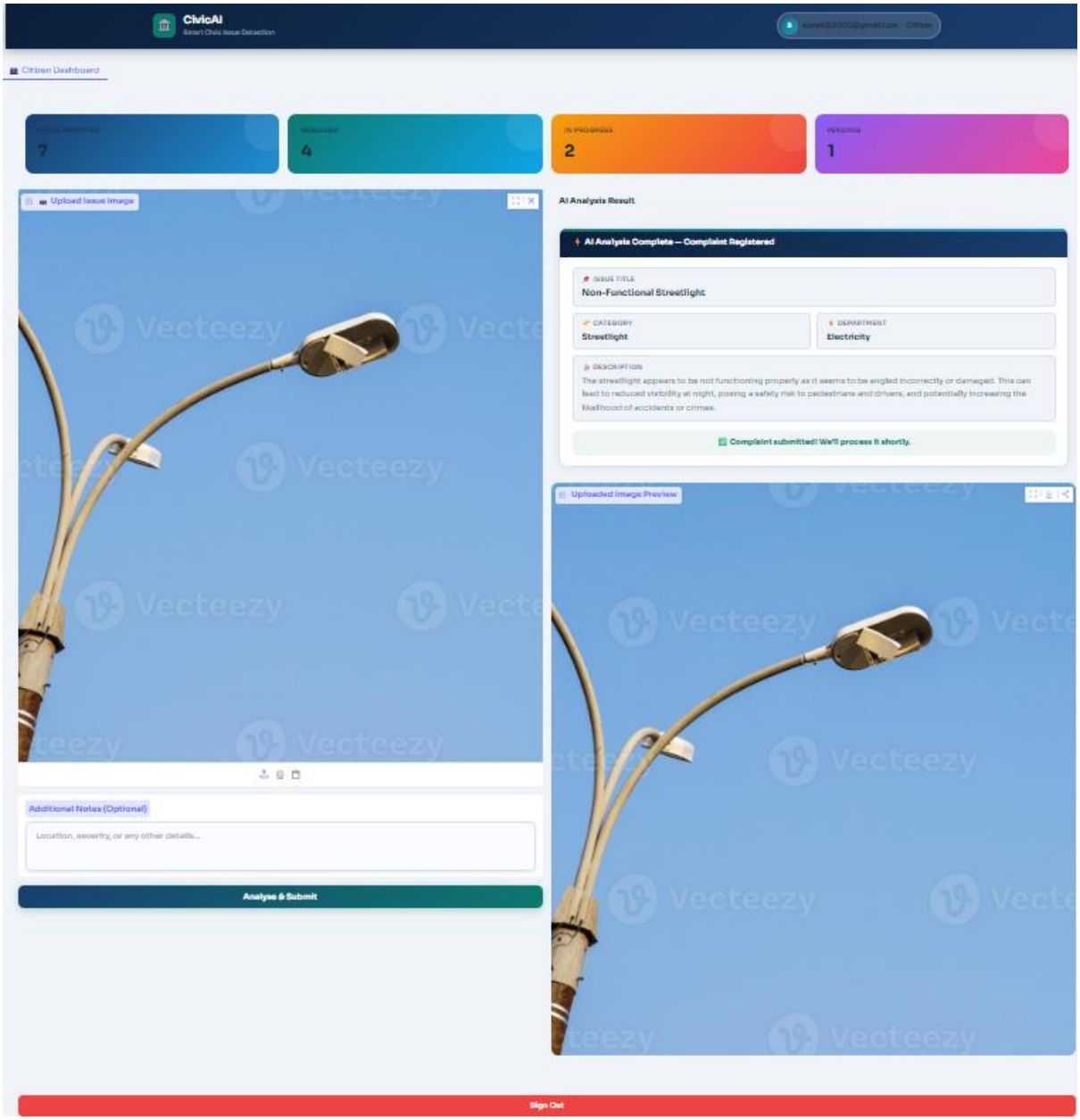
- Modern responsive interface using Gradio
- Role-based UI visibility
- Styled dashboards and cards

6. SCREENSHOTS

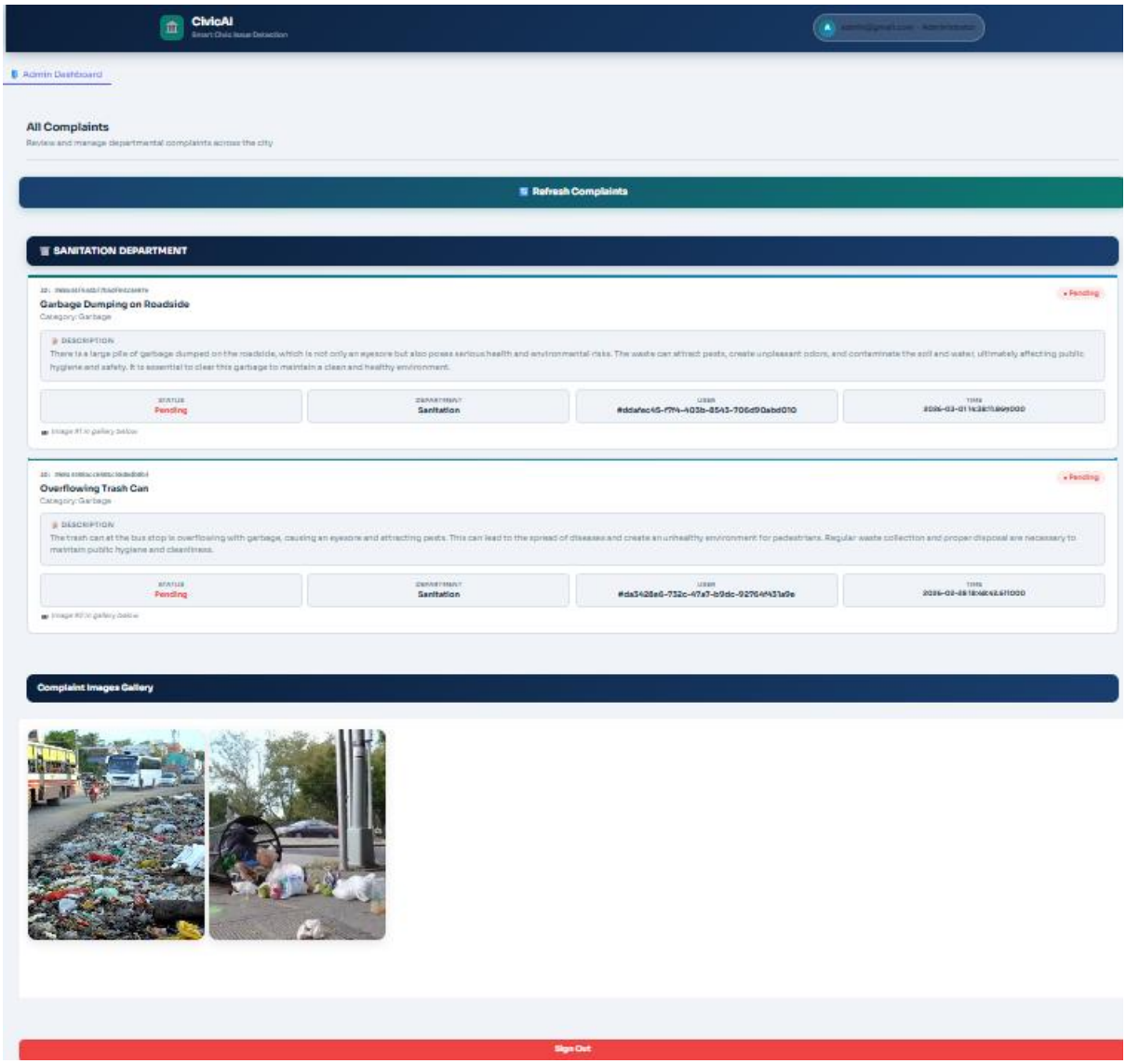
AUTHENTICATION PAGE



CITIZEN DASHBOARD


CivicAI
Smart Civic Issue Detection
Citizen Dashboard
7
4
IN PROGRESS
2
1
Upload Issue Image
AI Analysis Result
AI Analysis Complete — Complaint Registered
ISSUE TITLE
Non-Functional Streetlight
CATEGORY
Streetlight
DEPARTMENT
Electricity
DESCRIPTION
The streetlight appears to be not functioning properly as it seems to be angled incorrectly or damaged. This can lead to reduced visibility at night, posing a safety risk to pedestrians and drivers, and potentially increasing the likelihood of accidents or crimes.
Complaint submitted! We'll process it shortly.
Uploaded Image Preview
Additional Notes (Optional)
Location, severity, or any other details...
Analyze & Submit
Sign Out

ADMIN DASHBOARD



REOPSITORY LINK

https://github.com/Sonali-999/AI_CIVIC_ISSUE_REPORTER